

Learning the Effective PSF: A Neural PSF Field with Amortized Inference for DES Single-Epoch Imaging

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ABSTRACT

Errors in the modeled point-spread-function (PSF) are a leading source of systematic error in weak-lensing shear measurement. Per-exposure fitters (e.g., PSFEX, PIFF) represent the PSF as a sub-pixel surface-brightness profile that must be convolved with an assumed pixel response to predict an image. We instead target the *effective* PSF (the pixel-convolved point-source response): the quantity sampled data identify and the exact kernel for forward-modeling galaxies. A single neural network (multi-head attention over an exposure’s stars) predicts this effective PSF at any queried position and oversampling in one forward pass, *amortizing* the per-exposure fitting that PSFEX and PIFF repeat on every image. On 3,301 frozen test exposures of Dark Energy Survey single-epoch imaging, the model matches or exceeds PIFF and PSFEX on every reserved-star metric—size and ellipticity residuals, shape correlation, and per-pixel goodness of fit. In a star-anchored galaxy injection–recovery test on real data, it recovers galaxy *ellipticity*—the weak-lensing-critical quantity—without coherent bias and on par with PIFF, while recovered *sizes* run a few percent small. We trace this to the flux at which the PSF is queried: the model conditions on source flux and has learned, from training stars carrying faint unresolved neighbours, that fainter stars are more fractionally contaminated, so its predicted PSF narrows toward the clean PSF in the bright limit. Querying at that clean limit removes most of the size bias, returning the PSF concentration to truth in controlled simulations where contamination is the only flux-dependent effect. On real data the same query brings the predicted PSF size to within 1% of bright clean stars—from a $\sim 13\%$ second-moment excess at the median flux—and reduces the recovered galaxy-size bias toward the protocol floor. Color conditioning further removes a chromatic PSF-size systematic that the baselines retain, and the model stays accurate in star-sparse exposures where PIFF degrades sharply. Controlled simulations motivate two training choices: a *blend forward-model* loss that removes a size bias from unrecognized blends, and point-source-only attention, which fixes a spin-2 representation failure triggered by galaxy detections in the context.

Keywords: Weak gravitational lensing (1797) — Point spread function (1349) — Astronomical techniques (1684) — Convolutional neural networks (1938) — Sky surveys (1464)

1. INTRODUCTION

Weak gravitational lensing infers the matter distribution from coherent $\sim 1\%$ distortions of galaxy shapes. The measurement is limited by knowledge of the PSF: any error in the modeled PSF size or ellipticity propagates directly into the inferred shear. Survey pipelines model the PSF per exposure by fitting that exposure’s stars with a flexible model interpolated across the focal plane (PSFEX; E. Bertin 2011) (PIFF; M. Jarvis et al. 2021). This is robust but (i) discards information shared across exposures and nights, (ii) degrades when

few clean stars are available, and (iii) is purely empirical at the star positions: it does not exploit auxiliary information such as stellar color, which sets the chromatic PSF.

We take an amortized, learning-based approach. Rather than fit each exposure’s PSF from scratch, a single network learns, across many exposures, a *data-adaptive prior* over how the PSF looks and varies, and applies it to a new exposure in one forward pass, mapping that exposure’s stars (the *context*) to a continuous PSF field queryable at arbitrary position, color, flux,

and resolution. This pools information across the training set that no single-exposure fit can access, and in principle across the focal plane. The inferential target is the PSF at *star-free* positions, where galaxies are measured; predicting held-out stars is the training objective and the validation instrument. We train and evaluate on DES single-epoch data and benchmark against PIFF and PSFEX run by us under matched conditions.

A second difference is the *target* of the model. Traditional PSF models, including PSFEX (super-sampled pixels) and PIFF (a surface-brightness profile on the sky), represent the PSF as a continuous function on $\mathbb{R}^2$ at finer-than-pixel resolution, which must then be convolved with an assumed pixel-response kernel to predict what a detector records. We instead model the *effective* PSF (ePSF; J. Anderson & I. R. King 2000): the instrumental PSF already integrated over the pixel, i.e. the expected fraction of a point source’s light falling in each pixel as a function of the source’s sub-pixel position. The network maps a query position (and color, flux) directly to these expected pixel values, never representing the continuous PSF explicitly: the sense in which our PSF is *implicit*. Two grounds motivate this target. First, identifiability: from sampled data one cannot uniquely recover the sub-pixel surface brightness without extra assumptions, whereas the ePSF is exactly what the pixels constrain, the reason it underpins state-of-the-art point-source astrometry and photometry (J. Anderson & I. R. King 2000; T. R. Lauer 1999; T. I. Liaudat et al. 2023). Second, sufficiency: the ePSF as a function of source position is a sufficient statistic for forward-modeling *any* source: a galaxy’s expected pixel data is its surface brightness convolved with the ePSF, with no separate pixel step (G. M. Bernstein & M. Jarvis 2002), so the ePSF is precisely the kernel a shear pipeline needs. The only residual is discretizing that convolution on a finite grid, which our ability to render at arbitrary oversampling controls directly.

2. METHOD

Architecture—Each detected source is encoded from its pixel cutout (a learned image encoder applied to the scale-normalized stamp), its sky position (sinusoidal encoding), and scalar color and log-flux. A multi-head self-attention block (A. Vaswani et al. 2017) lets every query attend to the context sources; a coordinate decoder then renders the PSF at the query position and any oversampling factor, returning the pixel-convolved effective PSF used to convolve galaxy models. The decoder uses a polar, spin-2-aware angular parametrization, with FiLM modulation by the attended context features.

Inferential target (effective PSF)—The decoder outputs the effective PSF (J. Anderson & I. R. King 2000), the pixel-integrated point-source response, sampled at the query grid, optionally oversampled. Because the loss below compares predicted stamps to *observed* star pixels, the model regresses the observable by construction: it learns $\mathbb{E}[\text{pixel value} \mid \text{point source at } (x, y)]$ directly, rather than fitting a sub-pixel profile and assuming a pixel kernel. Identifiability aligns the three things that a non-identifiable model keeps separate: our training target, the quantity we model, and the quantity we validate all coincide, so reserved-star residuals (§4) directly test what we fit. A surface-brightness model is instead validated only through pixel- or moment-space summaries of its rendered output, the same observable space, so its extra sub-pixel degrees of freedom are never themselves constrained by the data. We score all three methods on moments measured in pixel space (model stamp versus star stamp), so no method is privileged by its internal representation. The same pixel-convolved output is the kernel used to forward-model galaxies in §4.4; oversampling the render reduces the discretization error of that convolution for compact sources.

Loss—Training minimizes an inverse-variance χ^2 between predicted and observed star cutouts, with the per-star amplitude solved in closed form. Star pixels are noisy realizations whose conditional mean is the effective PSF; a squared-error objective is minimized by the conditional expectation $\mathbb{E}[\text{pixel} \mid \text{source, context}]$, so with enough capacity and data the objective is minimized by that quantity, the identifiable effective PSF. The network thus targets only what the pixels constrain, rather than a finer-than-pixel profile the sampled data do not determine. We introduce a *blend forward-model* loss: rather than treat each clean star as isolated, we model its cutout as a generalized-least-squares sum of the target plus its detected neighbors, all rendered by the same decoder (§4.7). Detected galaxies near a target are handled by exclusion (default), masking, or a crude extended-component model; the three are within noise on real reserved stars.

Point-source context—The PSF should be informed only by point sources. We restrict the attention context to stellar detections; §4.7 shows this is necessary to avoid a spin-2 corruption from extended sources in the context.

Color conditioning—The decoder conditions on $g-i$ color, enabling a per-object chromatic PSF (differential chromatic refraction and wavelength-dependent seeing) that per-exposure empirical models cannot represent (§4.5).

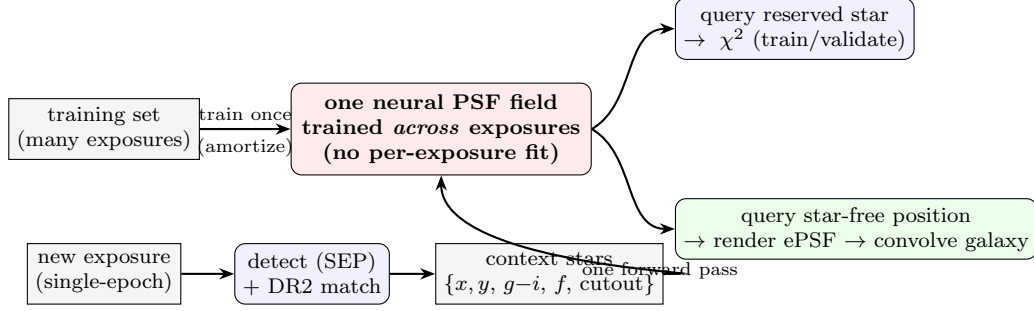


Figure 1. Overall workflow and amortization. A single network is trained once across many exposures (top) and then applied to a new exposure in one forward pass, with no per-exposure fitting: its detected stars (context) are mapped to a continuous PSF field (bottom). The *same* field is queried two ways: at held-out reserved stars, giving the training objective and validation, and at star-free positions, where the rendered (oversampled) effective PSF convolves galaxy models—the inferential target.

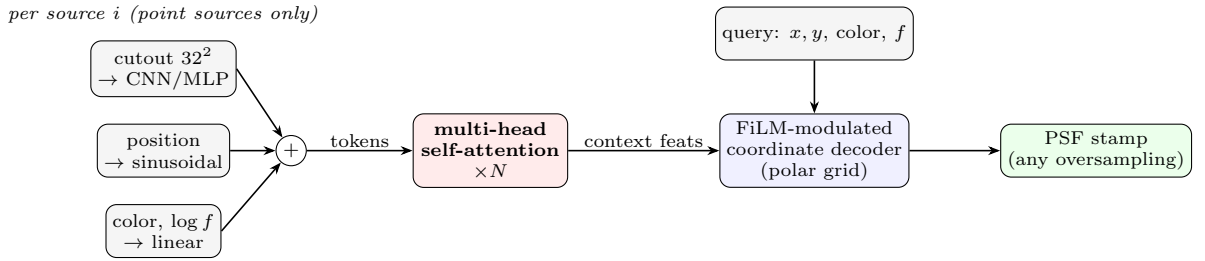


Figure 2. Network architecture. Each context source (restricted to point sources; §2) is encoded from its pixel cutout, sky position, and scalar color/log-flux into a token. Multi-head self-attention mixes the context; a FiLM-modulated coordinate decoder, conditioned on the attended features and the query (position, color, flux), renders the pixel-convolved PSF at any position and oversampling factor.

3. DATA

We use DES single-epoch FITS (Y5A1, CCD 31; amplifier B is dead, masked via MSK bit 8). Sources are detected with SEP and cross-matched to DES DR2 for colors, SPREAD_MODEL star/galaxy labels, and coadd object IDs. Splits are at the observing-night level by deterministic hash to prevent leakage (consecutive exposures share atmosphere), frozen in a committed manifest with $\sim 20\%$ of clean stars per test/validation exposure reserved for scoring only; the test split is 3,301 exposures across *grizY*. Reserved stars are excluded from every method’s fit/context and scored identically across methods on the same pixel grid.

Simulations—A matched pipeline renders Moffat PSF fields into the same data format, with FWHM and shear varying as second-order polynomials across the field. This variation lies within the polynomial spatial model that PIFF and PSFEX fit, so it does not handicap the baselines. The renderer optionally adds a chromatic dependence of FWHM on color, blended stars, and injected galaxy detections. Star density, galaxy density (3:1), and per-source S/N are tuned to match the real extraction. The simulation is the controlled instrument for the systematics studies in §4.7.

4. RESULTS

4.1. Reserved-star accuracy on real data

On 3,301 test exposures and 73,609 reserved stars, our model (no per-exposure fitting) matches or exceeds both baselines on every metric (Table 1). Every method is scored on the identical set of reserved stars.

4.2. Pixel-level and astrometric validation

Because the model regresses the effective PSF (the expected pixel values), the reserved-star quality of fit can be assessed directly in the modeled (pixel) space, the validation native to effective-PSF work (J. Anderson & I. R. King 2000). Two checks follow. First, the inverse-variance χ^2/dof is a noise-calibrated goodness of fit: at 1.04 (Table 1) the per-pixel residuals are consistent with photon and read noise, i.e. no unmodeled pixel-level structure, a statement available to a sub-pixel model only through its rendered output. The variance is the survey’s own inverse-weight (photon plus read noise) map, not a quantity tuned to the residuals, so $\chi^2 \approx 1$ is an independent calibration check rather than a fitted outcome. Second, the predicted ePSF centroid matches the star centroid to 0.010 px (2.6 mas) in median offset, versus 0.017 px (4.5 mas) for PIFF and 0.020 px (5.1 mas) for PSFEX (Table 2); all three centroids are

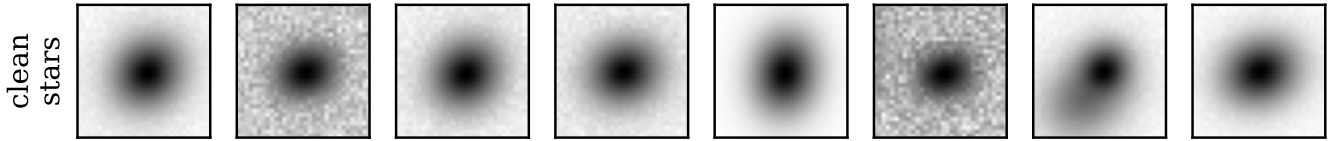
Example simulated cutouts (32×32 px, arcsinh stretch)

Figure 3. Example simulated clean-star cutouts (32×32 px, arcsinh stretch): the compact point sources that inform the PSF. Galaxy detections also populate the simulated fields but, under the point-source-context policy (§2, §4.7), are excluded from the attention context.

Table 1. Reserved-star residuals on the DES test split (3,301 exposures, 73,609 stars). T scatter is the robust standard deviation of $\delta T/T$; $|\delta e|$ are median absolute ellipticity residuals; corr is the model–star shape correlation. Best in bold.

Method	T scat	$ \delta e_1 $	$ \delta e_2 $	corr(e_1)	corr(e_2)	χ^2/dof
Neural PSF (ours)	0.0301	0.0151	0.0150	0.650	0.607	1.037
PIFF	0.0393	0.0171	0.0158	0.611	0.577	1.073
PSFEx	0.0371	0.0160	0.0154	0.645	0.604	1.047

True PSF ellipticity field (simulation)

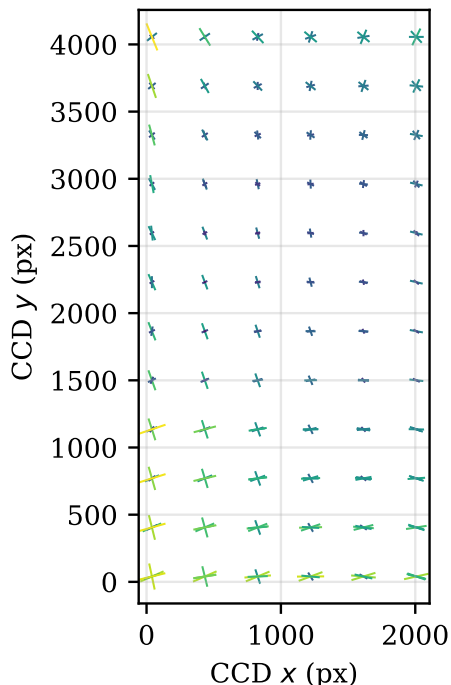


Figure 4. The true, spatially varying PSF ellipticity field in a simulated exposure (whisker length $\propto |e|$); the network must recover this at star-free positions.

measured with the same adaptive-moment routine on the same pixel grid, so the difference reflects the mod-

els, not a centroiding convention. This is a near $2\times$ improvement on a quantity the ePSF formalism is built to optimize.

Third, stacking the unit-normalized residual (observed star minus model, as a fraction of total light) over 306 reserved stars isolates any coherent shape error that second moments can miss (Figure 6). Our model leaves the flattest residual (RMS 2.6×10^{-4}); PIFF and PSFEx leave a visible negative core (3.5 and 4.4×10^{-4}), a higher-order mismatch in the central pixels that is consistent with their larger size residuals and is invisible to the size and ellipticity summaries alone.

4.3. Residual two-point statistics

Per-star residuals understate the impact on weak lensing, where the *spatially coherent* part of the PSF-modeling error sources additive shear bias. We therefore compute the standard ρ -statistics (B. Rowe 2010; M. Jarvis et al. 2016) of the reserved-star residuals (the auto- and cross-correlations of the residual ellipticity δe and the size-leakage term $e \delta T$), accumulated within exposures with TREECORR (Figure 7). On the same reserved stars used for Table 1, our model is at or below both per-exposure baselines on four of the five statistics: the median $|\xi_+|$ across angular scales is 3.9×10^{-5} (ρ_1), 5.2×10^{-5} (ρ_2), 2.5×10^{-6} (ρ_3), and 4.8×10^{-6} (ρ_4), against $\rho_1, \rho_2 \gtrsim 7 \times 10^{-5}$ and $\rho_3 \simeq 6 \times 10^{-5}$ for PIFF. The ρ_3 result, more than an order of magnitude below PIFF, reflects the model’s smaller and less spatially structured size residuals. The one exception

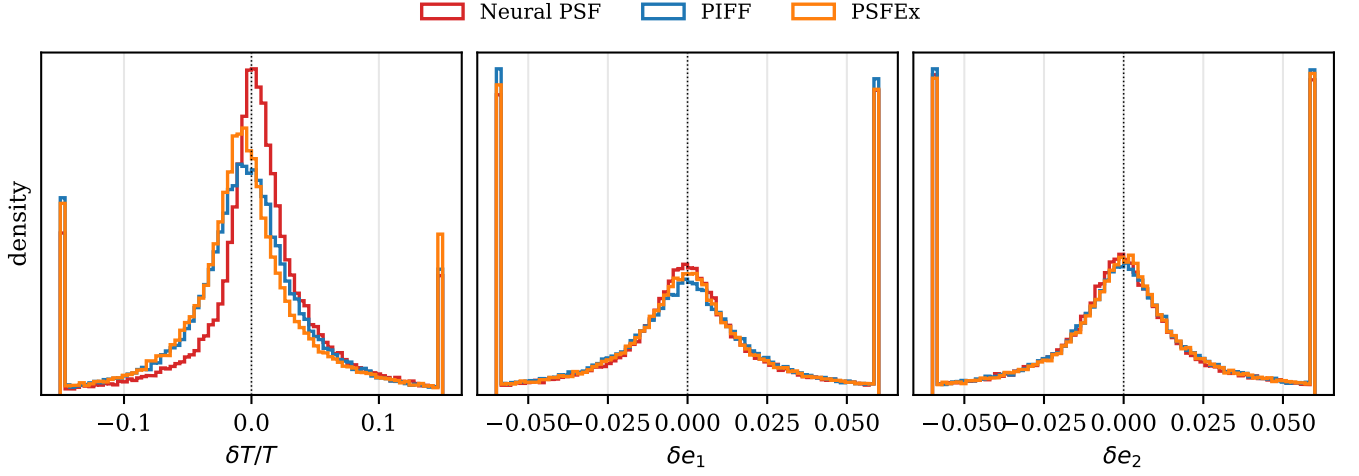


Figure 5. Reserved-star residual distributions on the DES test split. Our model (red) is the narrowest in size and ellipticity.

Table 2. Astrometric residual of the model PSF centroid relative to the star, on the reserved test stars. Per-axis scatter is the robust standard deviation; the offset is the median radius ($0.263''/\text{px}$). Best in bold.

Method	$\sigma(\delta x)$ [px]	$\sigma(\delta y)$ [px]	median $ \delta $ [mas]
Neural PSF (ours)	0.008	0.009	2.6
PIFF	0.015	0.013	4.5
PSFEx	0.015	0.017	5.1

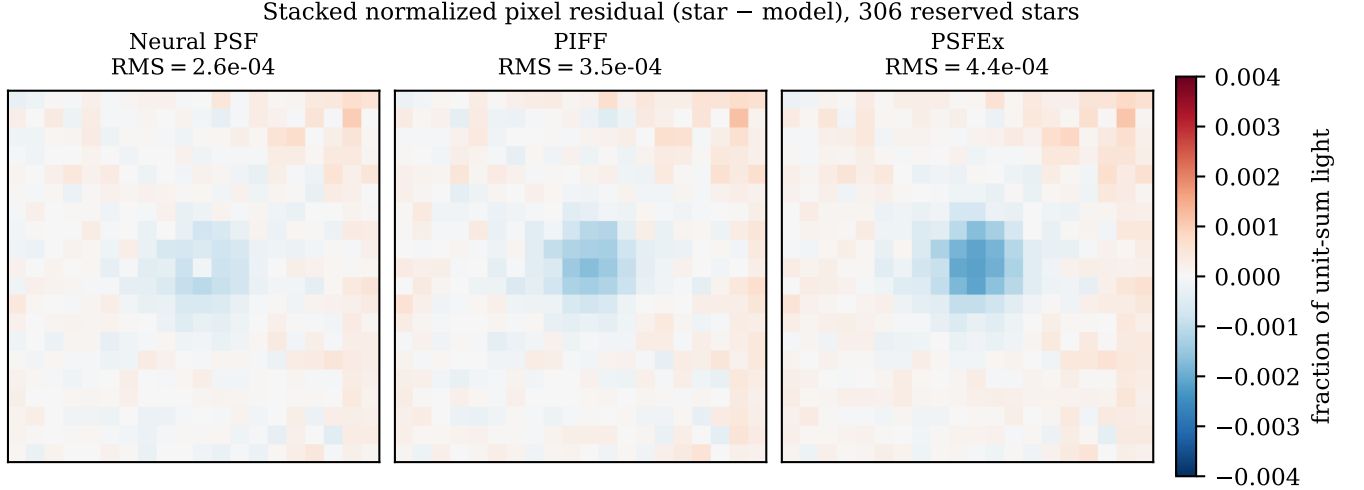


Figure 6. Stacked normalized pixel residual (observed star - model, unit-sum) over 306 reserved test stars, on a common scale. Our model leaves the flattest residual; the per-exposure baselines retain a coherent central core—a higher-order shape error that second-moment metrics do not capture (§4.2).

is $\rho_5 = \langle e(e\delta_T) \rangle$, where our model (5.4×10^{-5}) sits above the baselines ($\sim 10^{-5}$): a residual coupling between PSF shape and the size error that a weak-lensing analysis would carry in its additive-bias budget. Overall the coherent, lensing-relevant part of the residual is no larger than that of the per-exposure baselines, despite our model fitting no per-exposure free parameters.

4.4. Galaxy recovery on real data

We test the inferential goal directly. We inject GALSIM galaxies, convolved with a strictly isolated reserved star’s empirical PSF, at that star’s position; each method predicts the PSF there from the *other* stars and recovers the galaxy. Recovering with the true PSF (the anchor’s own image) sets the protocol floor: near-zero ellipticity bias and a residual -1.2% size bias—not a true zero, but the discretization floor of the recovery forward model—against which the predicted PSFs are scored. On 32 r -band test exposures (640 anchors; Table 3), all three methods recover galaxy *ellipticity* with low bias ($|\Delta e_1| \lesssim 0.016$): the pixel-convolved effective PSF carries the PSF ellipticity faithfully into the forward model, so no method imprints a coherent shape error on the recovered galaxies, the weak-lensing-critical property, which our model satisfies on par with the per-exposure baselines. The methods separate instead on recovered *size*: PIFF is slightly large ($+1.1\%$), while our recovered sizes are $\sim 8\%$ small. We trace this to the *flux at which the PSF is queried*. The model conditions on source flux, and—learning from training stars that carry faint unresolved neighbours—has encoded that fainter stars are more fractionally contaminated; its predicted PSF therefore narrows as the query flux rises, approaching the clean PSF in the bright limit. Queried at the median star flux, as in the recovery above, the predicted PSF is contamination-broadened, carrying $\sim 13\%$ too much second-moment area (and $\sim 2\%$ less encircled energy in the central few pixels) relative to bright, minimally-contaminated stars; this biases recovered galaxy sizes low. Queried instead at the clean high-flux limit, the predicted PSF size matches bright clean reserved stars to within 1% —a direct HSM-size residual of $+13.4\%$ at the median flux falling to $+0.0 \pm 0.4\%$ at 10^6 (Table 4)—and the recovered galaxy-size bias collapses correspondingly: on 32 matched held-out exposures ($n = 640$ anchors) the de-confounded bias (implicit minus the true-PSF recovery) falls from -4.6% at the median flux to -0.9% at the clean flux, i.e. essentially to the protocol floor. The recovery is robust across all production seeds (the predicted PSF size returns to within 1% of bright clean stars at the clean flux for every seed, though the median-flux deficit mag-

Table 3. Star-anchored galaxy injection–recovery (32 r -band test exposures, 640 anchors). The true-PSF recovery sets the protocol floor. All methods recover ellipticity with low bias; they separate on recovered size. Rendering our model’s PSF at the median query flux leaves a size deficit (contamination-broadening, §4.4); at the clean high-flux limit it is largely removed (Table 4).

PSF	Size bias	Δe_1	Δe_2
True PSF	-1.3%	-0.009	$+0.004$
Neural PSF (median flux)	-5.8%	-0.006	$+0.005$
Neural PSF (clean flux)	-2.2%	-0.011	$+0.005$
PIFF	$+1.3\%$	-0.016	$+0.003$

nitude is itself somewhat seed-dependent). The clean window is broad: the size residual is flat from 5×10^5 to 6×10^6 , brighter-fatter contributing only $+0.2\%$ across that decade. Controlled simulations confirm the mechanism against *known* truth, with no brighter-fatter to confound: the same flux query drives the learned PSF concentration from $\delta EE = -0.0078$ at the median flux to -0.0014 at the clean limit (§4.7), i.e. essentially onto the true PSF. That the model recovers the clean PSF purely through its learned flux dependence is independent confirmation that faint-neighbour contamination of the training stars is the cause. The size bias is thus largely a query-flux effect rather than an irreducible limitation; recovered ellipticity, the consequential quantity for shear, is unbiased throughout.

4.5. Chromatic PSF correction

In a chromatic simulation (FWHM varies $-3\%/mag$ of $g-i$), color conditioning reduces the residual chromatic size slope from $-0.06/mag$ (PIFF, PSFEx, and our own zero-color ablation) to $+0.004/mag$, a $\sim 15\times$ reduction. On real r -band data the paired color-versus-zero-color slope difference is $+0.0024/mag$ (CI $[+0.0016, +0.0030]$); the baselines retain the chromatic slope because a per-exposure empirical PSF has no access to per-star color.

4.6. Sample efficiency

Restricting every method to k randomly chosen fit stars, our model is approximately flat in k (size scatter $0.078 \rightarrow 0.056$, $\chi^2 \sim 1.1$ from $k = 5$ to $k = 100$), while PIFF collapses below $k \approx 25$ ($\chi^2 = 5.1$, scatter 0.18 at $k = 5$). Amortization transfers information from the training set to sparsely-populated exposures.

PSF residual rho-statistics (reserved stars, all bands)

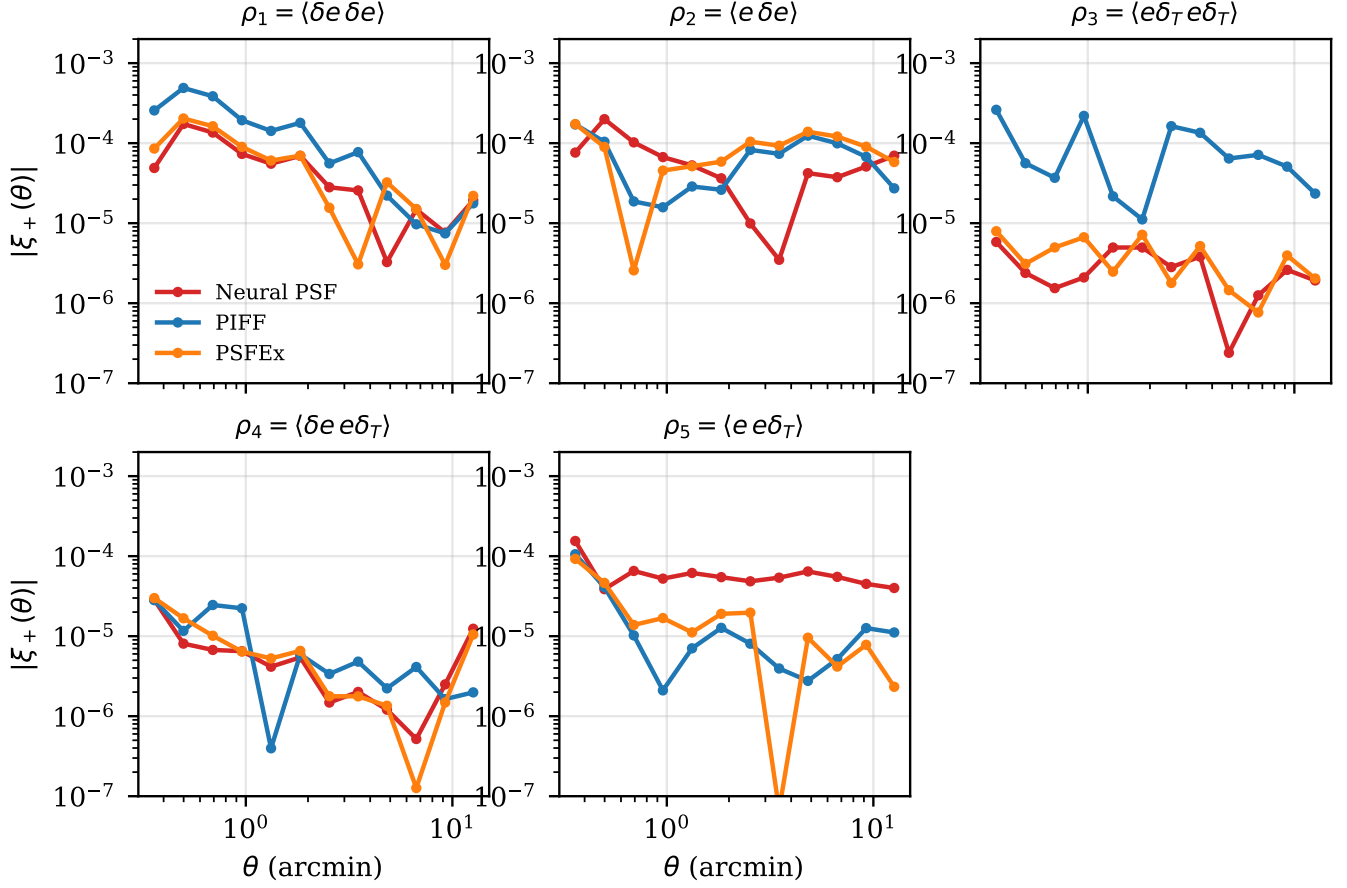


Figure 7. PSF residual ρ -statistics on reserved stars, per method. Our model (red) is at or below both baselines on ρ_1 – ρ_4 ; only ρ_5 (the shape–size coupling) is modestly larger (§4.3).

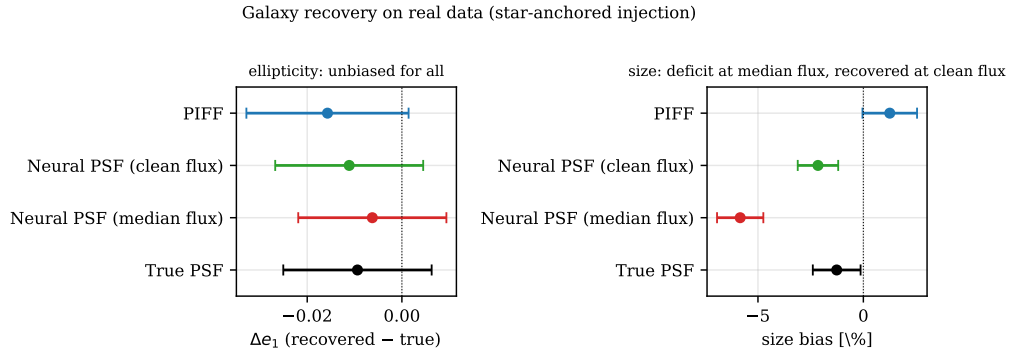


Figure 8. Galaxy recovery on real data (star-anchored injection), median $\pm$ bootstrap 95% CI per PSF. Left: recovered ellipticity is low-bias ($|\Delta e_1| \lesssim 0.016$) for all three methods. Right: recovered *size*. Queried at the median flux our model runs small (contamination-broadening of the predicted PSF); queried at the clean high-flux limit it recovers to near the true-PSF floor (§4.4). PIFF is shown for reference.

Table 4. Predicted PSF second-moment size residual $\delta T/T$ (HSM adaptive moments) against bright clean reserved stars, as a function of the flux at which the PSF is queried (50 held-out r -band exposures, $n = 260$ stars, $S/N \geq 200$; mean $\pm$ s.e.m.). The under-concentration is a query-flux effect: at the median star flux the predicted PSF is contamination-broadened; at the clean high-flux limit it matches bright clean stars, and the window is broad and flat ($\delta T/T$ stays within 1% from 5×10^5 to 6×10^6 , brighter-fatter contributing $+0.2\%$ across that decade).

Query flux (ADU)	$\delta T/T$
median star flux	$+13.4 \pm 0.6\%$
10^5	$+6.7 \pm 0.5\%$
5×10^5	$+0.2 \pm 0.4\%$
10^6	$+0.0 \pm 0.4\%$

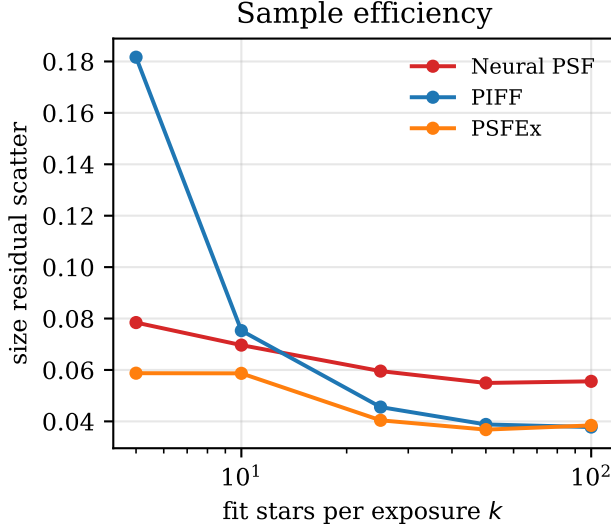


Figure 9. Size-residual scatter versus the number of fit stars k . Our model is nearly flat in k ; PIFF degrades sharply below $k \approx 25$.

4.7. Simulation studies

Architecture ladder—On the star-free truth grid, χ^2/dof improves along additive $\rightarrow$ FiLM $\rightarrow$ diagonal $\rightarrow$ polar ($8.9 \rightarrow 4.96 \rightarrow 4.41$; PIFF 2.67; verified floor 1.0), recovering both ellipticity components.

Blend forward-model—On a blended simulation, single-star training leaves a $+4.8\%$ size bias on the star-free truth grid; the blend forward-model loss reduces it to $+0.7\%$ ($\approx 7\times$).

The size deficit reproduces in simulation and is a query-flux effect—Adding the faint-neighbour contamination of §4.4 to the training stars reproduces the size deficit in simulation, where the truth PSF is known exactly: the learned PSF is under-concentrated by $\delta EE(r \leq 2\text{px}) = -0.0078$ relative to truth at the median query flux, and injected galaxies are recovered -2.4% small (the simulation analogue of the real deficit, after differencing against recovery through the known clean PSF to cancel the protocol floor). Because the simulation has no brighter-fatter, the model’s only flux-dependent effect is the contamination it learned, and querying its PSF at the clean high-flux limit—with no cleaning at all—drives δEE from -0.0078 to -0.0014 , i.e. essentially onto truth. This is the controlled, truth-anchored counterpart of the real-data flux query in §4.4, confirming against known truth that the model’s bright-limit PSF is the clean one and that the deficit is the contamination the flux-conditioned model has implicitly learned to separate. This survives a deliberately adversarial test of the real selection: when the detectability threshold is made to rise with host brightness—so that bright “clean” survivors retain *more* masked contamination, as the real SEP filter allows—the flux query still recovers truth ($\delta EE = -0.0059 \rightarrow -0.0005$ from median to clean flux). The query relies only on bright stars being *fractionally* cleaner, which holds even when their retained neighbours are absolutely brighter, so the recovery is not an artifact of assuming bright stars are perfectly clean.

Galaxy-context corrupts PSF ellipticity—and the fix—Injecting galaxy detections into the simulation, a model that attends to *all* detections (including galaxies) suffers a clean, reproducible failure: it cannot represent one spin-2 ellipticity component [$\text{corr}(e_1) = 0.05$ under the polar parametrization; the failure flips to e_2 under a Cartesian parametrization, identifying a representation degeneracy rather than noise], while recovering the other perfectly. We localize the cause by ablation: it is *not* data volume, per-source S/N , galaxy size or color, or the χ^2 cap—the model is treating extended galaxy cutouts as PSF evidence. Restricting the attention context to point sources recovers *both* components completely [$\text{corr}(e_1, e_2) = 0.98$, matching the galaxy-free ideal]; a coordinate-system change does not. Galaxies are clearly distinguishable (cutout second moment $T \approx 33$ versus 15px^2 for stars), so this is the model failing to use available information, fixed by the principled

Table 5. Truth-grid PSF recovery (4 matched 120-epoch seeds): seed-noise scatter for a single model, EMA, and a 4-seed ensemble.

	$\sigma(\delta e_1)$	$\sigma(\delta e_2)$	$\sigma(\delta T/T)$
single model	0.0094	0.0099	0.0175
EMA (1×)	0.0092	0.0096	0.0167
ensemble (4×)	0.0073	0.0075	0.0154

restriction—which we adopt as the production policy—that only point sources inform a PSF. Excluding galaxies from the context costs nothing measurable on the real test set: reserved-star residuals are unchanged from the all-detections ablation (size scatter 0.031 versus 0.030; identical median ellipticity error and shape correlation), so the policy removes the simulation failure mode without trading away real-world performance.

Seed regularization: EMA versus ensembling—Because the field is learned once and applied without per-exposure refitting, it carries seed-to-seed prediction noise. On the star-free truth grid we compare a single model, an exponential moving average (EMA) of one model’s weights, and a 4-seed deep ensemble (averaging predicted stamps), each over matched 120-epoch runs (Table 5). EMA gives a small, consistent gain (T -scatter -5% , ellipticity $\lesssim 3\%$); the ensemble is much larger (-12% size, -24% ellipticity). EMA is therefore *not* a substitute for ensembling: it captures only a fraction of the benefit. The per-seed aggregate scatter is itself small (the model is seed-stable), but the per-position predictions are diverse, which only ensembling exploits. On real reserved stars the ensemble improves the metrics by only $\sim 1\%$: that residual is measurement-noise- and blend-limited (corr ~ 0.66 , not the truth-grid 0.99), so it masks the model-error reduction that ensembling delivers at the star-free positions where galaxies are measured.

5. DISCUSSION

Our amortized approach trades a per-exposure fit for a single trained model. Benefits: no test-time fitting, robustness in star-sparse exposures, conditioning on color (chromatic PSF) and flux (brighter-fatter), and a directly forward-modelable output: we predict the pixel-convolved effective PSF, the identifiable quantity the data constrain and the exact kernel a galaxy model is convolved with (§1), so the PSF feeds a shear pipeline with no intervening pixel-deconvolution assumption. Caveats: (i) unlike per-exposure fitters,

our approach requires many exposures to learn the field (in simulation, ellipticity recovery is unstable below a few thousand exposures; real DES is well above this); (ii) the simulation’s idealized PSF (Moffat, low-order polynomial field) favors the per-exposure baselines, so absolute-accuracy comparisons rest on real data; and (iii) the galaxy-context finding argues that a PSF estimator’s context should be gated to point sources, which we adopt. (iv) The model’s predicted PSF *size depends on the flux at which it is queried*: it conditions on flux and has learned, from training stars that carry faint unresolved neighbours, that fainter stars are more fractionally contaminated and hence broader, so its predicted PSF narrows toward the clean PSF as the query flux rises. Queried at the median star flux—which a naive galaxy-recovery render uses—the predicted PSF is contamination-broadened (a second-moment size $\sim 13\%$ large, $\sim 2\%$ low in central encircled energy, relative to bright clean stars) and biases recovered galaxy sizes a few percent low; queried at the clean high-flux limit it matches bright clean stars to 1%, the size residual falling from $+13\%$ to 0% across the flux range, and the galaxy-size bias largely vanishes (§4.4). The flux-size trend is visible directly in the reserved-star metric (Fig. 14), $\delta T/T$ rising at faint magnitudes. Controlled simulations with known truth confirm the mechanism and that the clean-flux PSF is the true one (δEE from -0.0078 to -0.0014 ; §4.7), and the effect is absent at any decoder capacity in clean simulations, ruling out a generic spectral-bias limitation of the decoder. We identify faint-neighbour contamination of the nominally-clean training stars as the cause, and the practical prescription—render the inferential PSF at the clean-flux limit—removes most of it, the main residual systematic to control before a shear analysis. As a principled cross-check we also fit the contamination explicitly, by a Monte Carlo expectation maximization (A. P. Dempster et al. 1977; G. C. G. Wei & M. A. Tanner 1990) that infers each star’s contaminant posterior (with the contamination rate inferred jointly across stars rather than assumed) and retraining the field against the cleaned stars; in controlled simulations this reaches the same truth as the flux query, to within the seed scatter, confirming the mechanism but not improving on the simpler query, which we therefore adopt as the operative correction.

6. CONCLUSION

A single attention-based continuous PSF field, trained across DES single-epoch exposures and applied with no per-exposure fitting, matches or exceeds PIFF and PSFEX on real reserved stars and recovers galaxy ellipticity without coherent bias (on par with PIFF) on

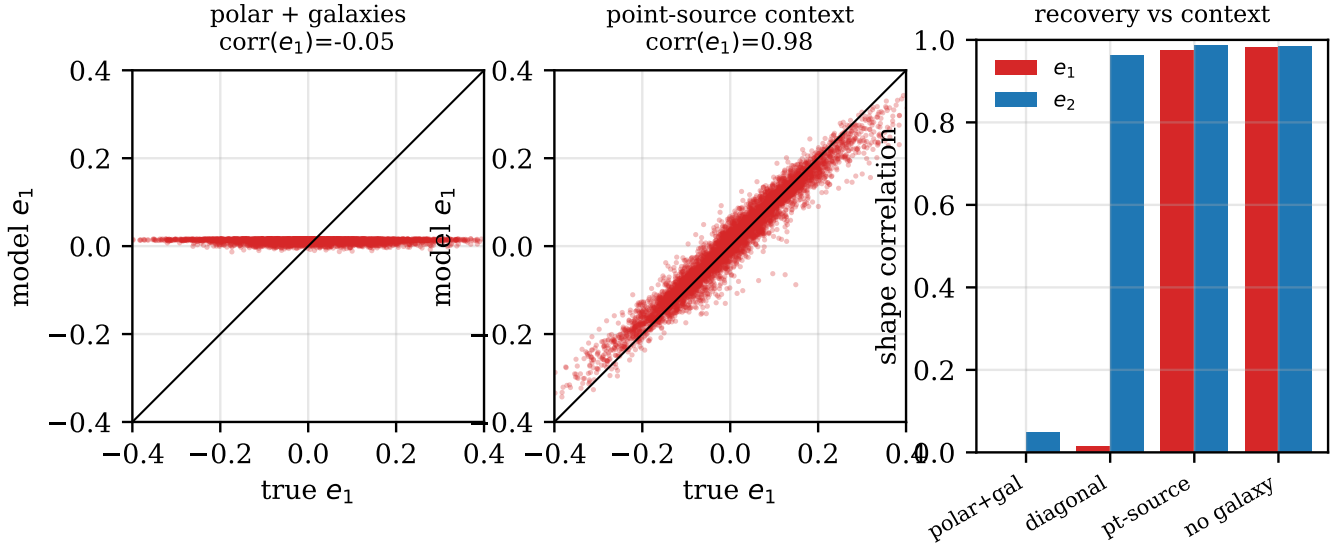


Figure 10. Galaxy detections in the attention context corrupt the recovered PSF ellipticity. Left: with galaxies in context, model e_1 collapses to a constant ($\text{corr} = -0.04$). Middle: restricting the context to point sources recovers it ($\text{corr} = 0.98$). Right: shape correlations across configurations—point-source context and galaxy-free both recover *both* components; a coordinate change (diagonal) does not.

a star-anchored injection test, while additionally correcting a chromatic systematic and remaining robust in star-sparse exposures. The one residual—recovered galaxy sizes a few percent small—we trace to the flux at which the PSF is queried: the flux-conditioned model has learned that fainter stars are more contaminated, so rendering the PSF at the clean high-flux limit removes most of the bias. Controlled simulations motivate two further training choices: a blend forward-model loss and point-source-only attention context. The latter follows from diagnosing a spin-2 representation failure triggered by galaxy detections in context.

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Baselines: PIFF (PixelGrid, second-order BasisPolynomial, 4σ rejection) and PSFEX (PSF_SAMPLING 0.5, 45×45 , PSFVAR_DEGREES 2; rendered via GALSIM DES_PSFEX, method=’no_pixel’).

Software: Astropy (Astropy Collaboration 2022), GalSim (B. T. P. Rowe et al. 2015), SEP (K. Barbary 2016), PyTorch (A. Paszke et al. 2019), PIFF (M. Jarvis et al. 2021), PSFEX (E. Bertin 2011)

APPENDIX

A. REAL-DATA STAR SELECTION AND PSF EXAMPLES

Figure 11 shows a region of a real DES single-epoch r -band exposure with the source selection overlaid: clean stars used for fitting and scoring (green), held-out reserved stars (blue), and additional context-only stellar detections (orange). Galaxies (the extended sources, e.g. the bright object near the center) are detected but excluded from the attention context (§2). Figure 12 compares, for a single high-S/N reserved test star, the observed stamp against the PSFEX, PIFF, and Neural PSF models evaluated at that position (each method having seen only the non-reserved stars). All three reproduce the stellar profile; PIFF’s per-exposure interpolation is visibly noisier in the wings.

B. BENCHMARK VALIDATION DIAGNOSTICS

For comparison with standard per-exposure PSF validation (cf. M. Jarvis et al. 2021), we report the same diagnostics.

Table 6 breaks the metrics down by band. Our model has the smallest size scatter and median ellipticity error in every band.

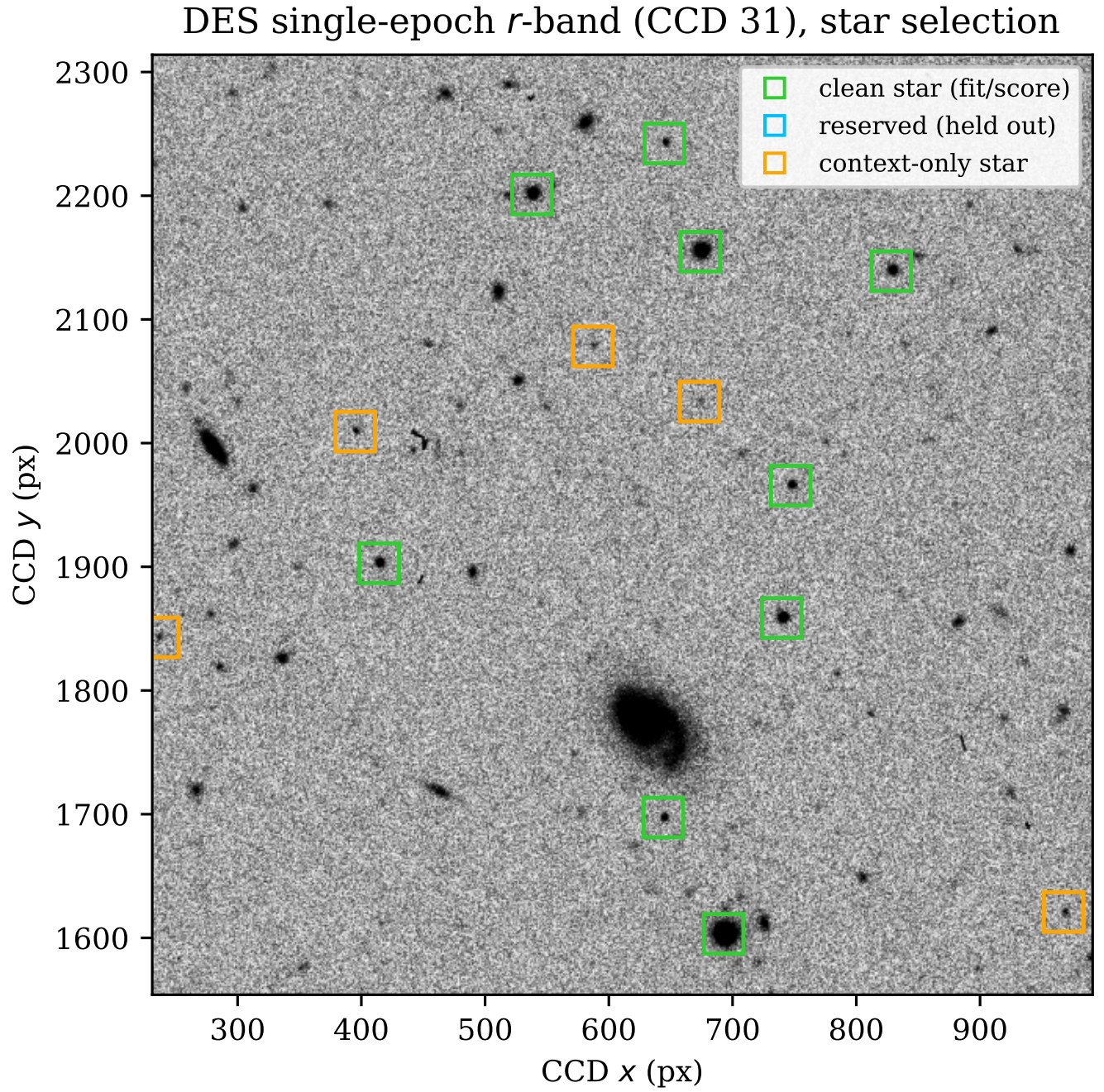


Figure 11. Star selection on a real DES exposure. Green: clean stars (fit and score). Blue: reserved (held out for evaluation). Orange: context-only stellar detections. Extended sources (galaxies) are excluded from the attention context.

Same reserved test star: observed, modeled PSF, and residual

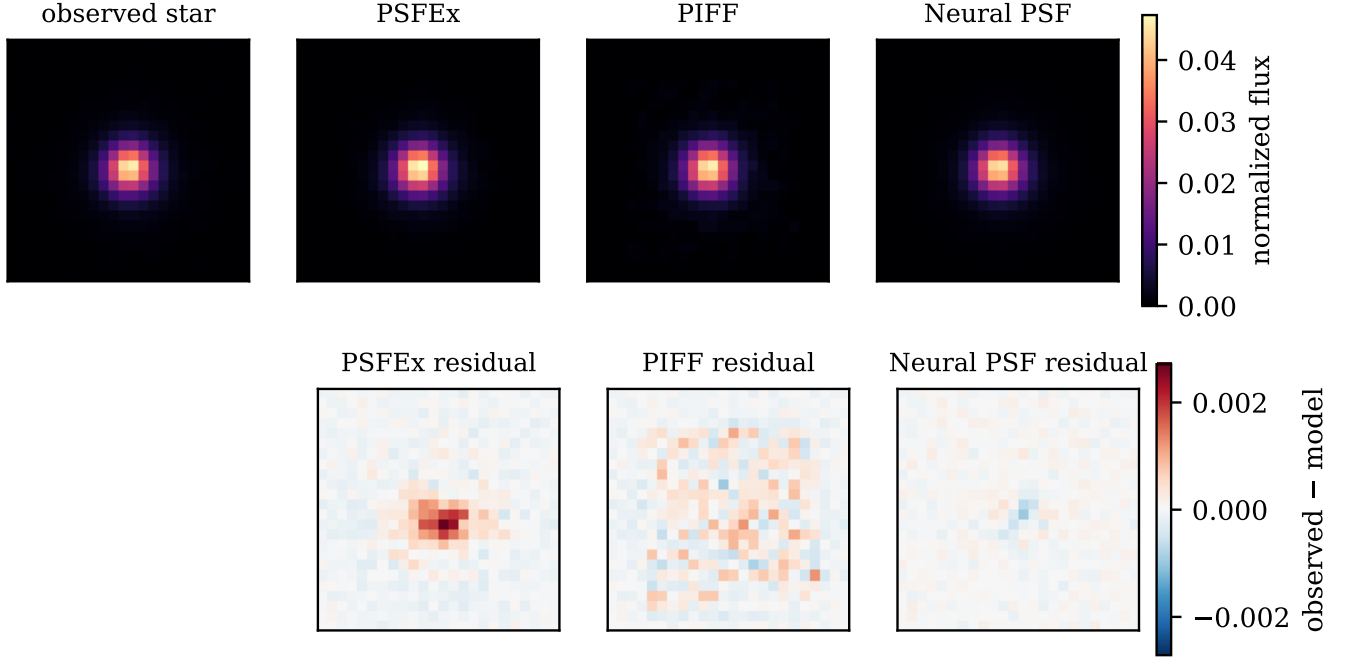


Figure 12. Top: the same reserved test star (left, observed) and the PSF predicted at its position by PSFEx, PIFF, and Neural PSF, on a common linear stretch. Bottom: the residual (observed $-$ model) for each method, on a common diverging scale. Each model was fit/conditioned on the non-reserved stars only.

Table 6. Reserved-star residuals by band (test split). T scatter is the robust standard deviation of $\delta T/T$; $|\delta e_1|$ is the median absolute e_1 residual; corr is the model-star e_1 correlation. Best per band in bold.

Band	Method	T scat	$ \delta e_1 $	corr(e_1)
g	Neural PSF (ours)	0.0331	0.0164	0.475
	PIFF	0.0439	0.0188	0.448
	PSFEx	0.0417	0.0173	0.473
r	Neural PSF (ours)	0.0291	0.0144	0.576
	PIFF	0.0366	0.0164	0.535
	PSFEx	0.0340	0.0151	0.572
i	Neural PSF (ours)	0.0274	0.0136	0.643
	PIFF	0.0342	0.0155	0.605
	PSFEx	0.0339	0.0144	0.637
z	Neural PSF (ours)	0.0272	0.0140	0.601
	PIFF	0.0387	0.0160	0.503
	PSFEx	0.0359	0.0152	0.589
Y	Neural PSF (ours)	0.0391	0.0210	0.568
	PIFF	0.0513	0.0226	0.523
	PSFEx	0.0449	0.0221	0.550

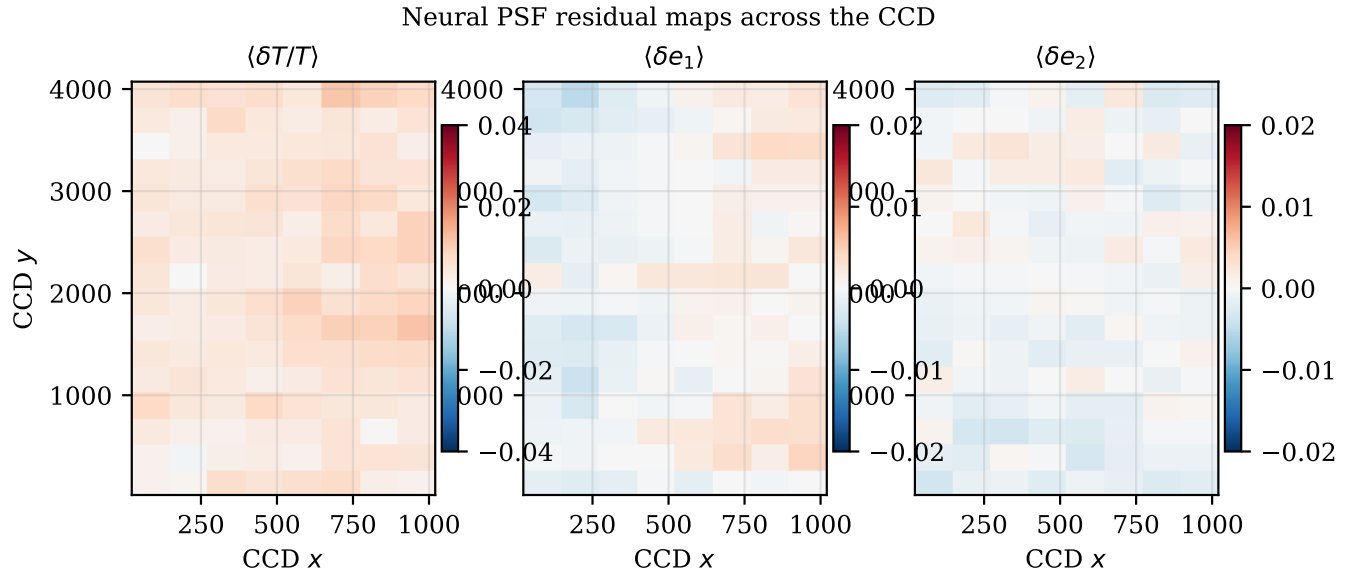


Figure 13. Median Neural PSF residuals binned across the CCD (cf. PIFF Fig. 9).

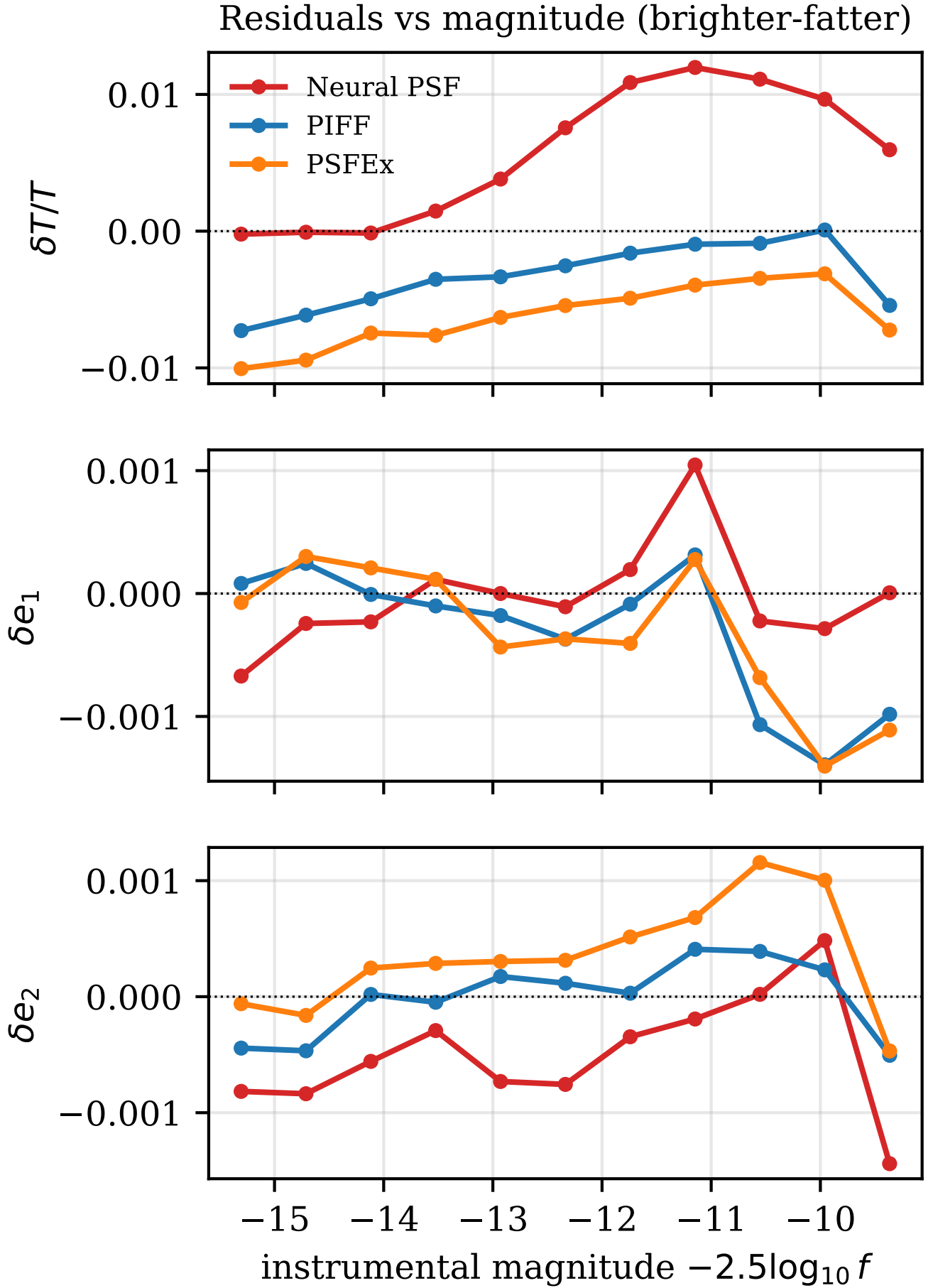


Figure 14. Residuals versus instrumental magnitude (cf. PIFF Fig. 10). Our model (red) carries a magnitude-dependent size residual peaking at $\pm 1.2\%$ (85).

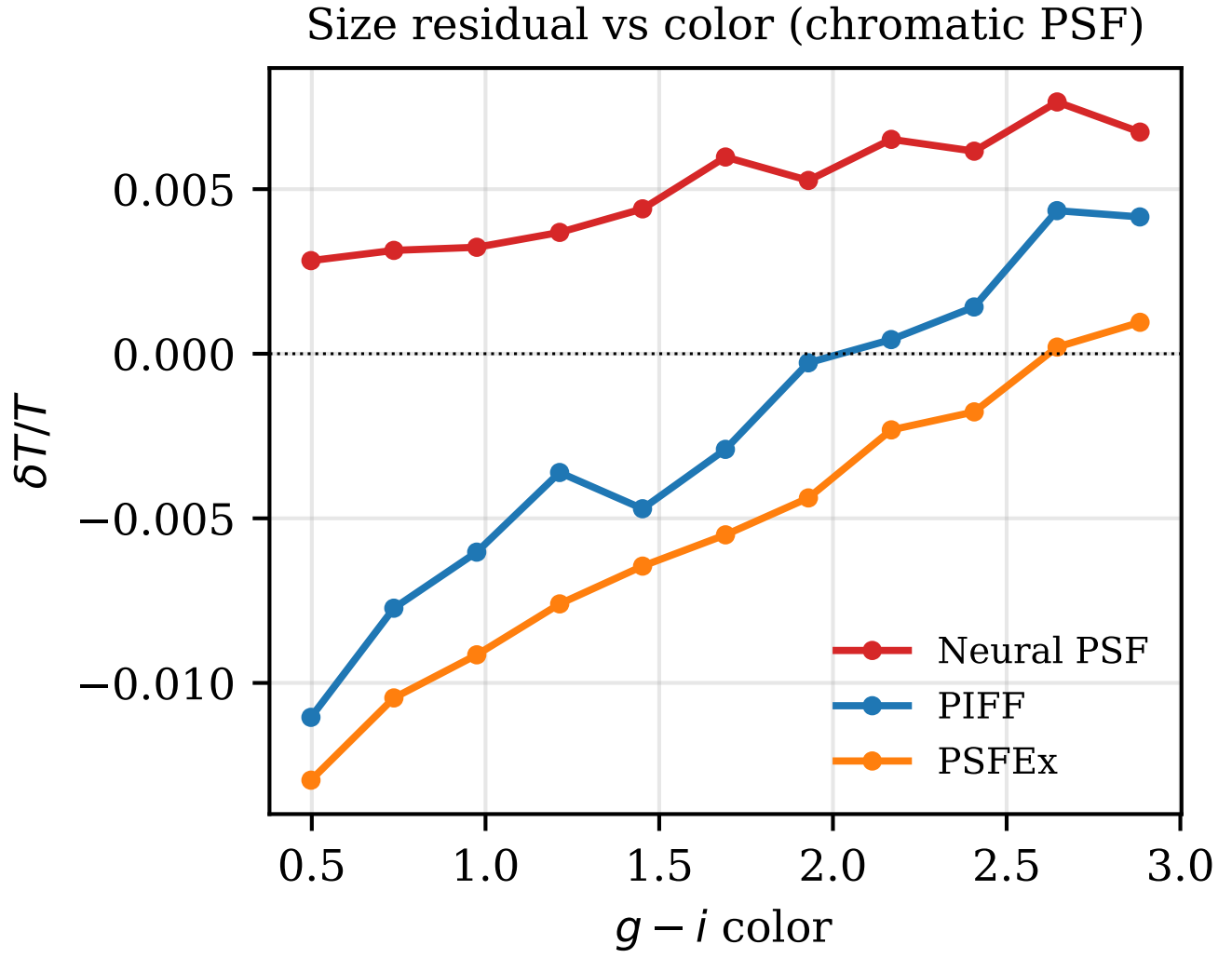


Figure 15. Size residual versus $g - i$ color (cf. PIFF Fig. 11); color conditioning keeps our chromatic residual flat.

Residual PSF ellipticity whiskers

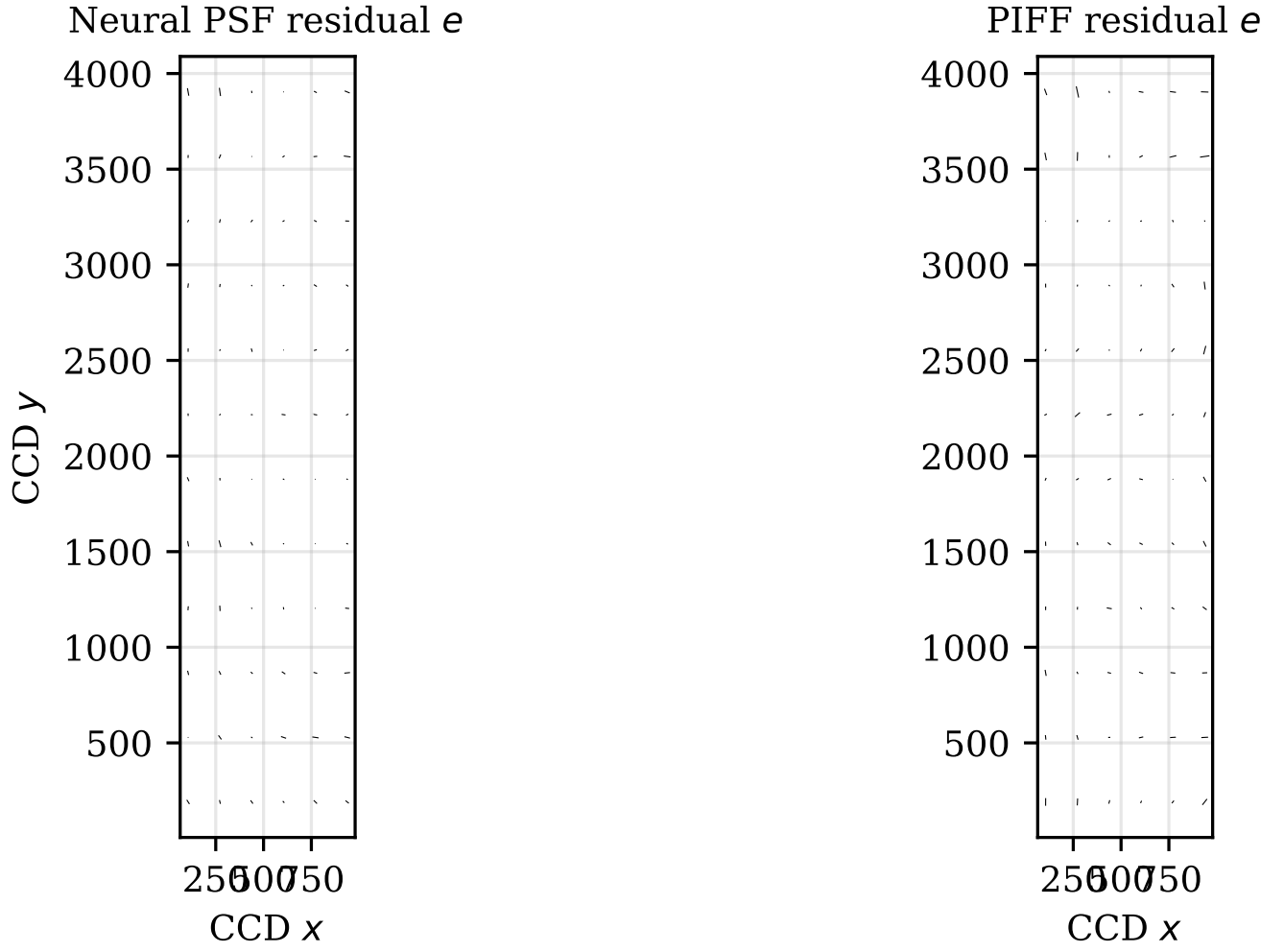


Figure 16. Residual PSF-ellipticity whiskers across the CCD, Neural PSF and PIFF (cf. PIFF Fig. 2).

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